Portfolio Risk–Return Analysis of Selected JSE Stocks

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Objective: Evaluate historical performance and identify the optimal risk-adjusted portfolio allocation.

Executive Summary

This analysis evaluates five JSE-listed equities between 2018 and 2025 to assess return, volatility, and diversification characteristics.

Key objectives:

- Compare individual stock performance
- Measure risk using standard deviation
- Analyse correlations between assets
- Simulate portfolio allocations
- Identify the portfolio with the highest Sharpe ratio

The goal is to determine whether diversification improves risk-adjusted returns relative to holding a single asset.

Summary of Findings

- SHP.JO delivered the strongest risk-adjusted performance
- SOL.JO exhibited highly disproportionate volatility
- Optimal portfolio heavily weighs on SHP.JO due to its superior risk-adjusted profile and allocates minimal capital to SOL.JO given its high risk and low return contribution.
- Max Sharpe portfolio: 18.3% return, 24% volatility

Business Problem

Investors must balance expected return against volatility.
Holding a single high-return stock may expose an investor to excessive risk.

This analysis explores whether combining assets across sectors (financials, retail, resources, and global tech exposure) improves portfolio stability and maximises risk-adjusted performance.

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Data Overview

The dataset contains daily adjusted closing prices for each selected stock. Adjusted prices account for dividends and stock splits, making them suitable for total return analysis.

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Ticker	AGL.JO	FSR.JO	PRX.JO	SHP.JO	SOL.JO
count	1326.000000	1326.000000	1326.000000	1326.000000	1326.000000
mean	52442.507444	6108.259679	60267.431069	20022.162363	2913.694329
std	11178.611196	1135.642249	11502.200197	5977.685725	8509.480432
min	20602.863281	3378.772461	31634.068359	10012.396484	2185.266846
25%	42851.330078	5477.398071	53063.999023	13898.250732	14096.030518
50%	53589.796875	6264.215332	60895.279297	21259.180664	23074.265625
75%	59503.603539	6737.963518	67378.70172	24481.319824	27664.897461
max	81421.523438	8681.484375	88903.789062	31443.160156	42988.234375

Exploratory Data Analysis (EDA)

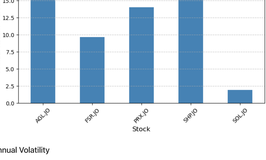
Ticker	AGL.JO	FSR.JO	PRX.JO	SHP.JO	SOL.JO
count	1325.000000	1325.000000	1325.000000	1325.000000	1325.000000
mean	0.000705	0.000381	0.000554	0.000846	0.000076
std	0.026036	0.021017	0.025502	0.018798	0.044381
min	-0.133194	-0.142791	-0.173925	-0.099922	-0.465627
25%	-0.013320	-0.010098	-0.012099	-0.009392	-0.019197
50%	0.000740	0.000656	-0.000209	-0.000035	-0.001077
75%	0.014768	0.011068	0.012680	0.009765	0.016343
max	0.189202	0.137758	0.210721	0.136302	0.534834

Average Annual Returns

The average annual returns indicate differences in long-term performance across the selected JSE stocks.

- **SHP.JO (21.33%)** delivered the highest average annual return, suggesting strong historical growth over the selected period.
- **AGL.JO (17.78%)** demonstrated solid performance, likely reflecting its exposure to global commodity cycles.
- **PRX.JO (13.97%)** showed moderate growth that is consistent with its international tech exposure.
- **FSR.JO (9.61%)** produced steady but comparatively lower returns, typical of a large financial institution.
- **SOL.JO (1.91%)** underperformed significantly compared to other assets. This indicates weak historical growth during the selected period.

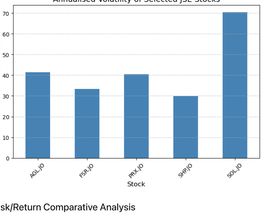
Retail and resource exposure from SHP.JO and AGL.JO appear to have generated the strongest returns, while energy (SOL.JO) contributed the least.



Annual Volatility

- **SOL.JO (70.45%)** shows extremely high volatility, indicating significant price fluctuations and high risk exposure.
- **AGL.JO (41.33%) and PRX.JO (40.48%)** show relatively high risk, reflecting sensitivity to global commodity and tech market movements.
- **FSR.JO (33.36%)** reflects moderate risk, consistent with the exposure in large-cap financial sectors.
- **SHP.JO (29.84%)** exhibits stable price behaviour, as shown in its lower volatility.

Financial and retail exposures provide lower risk profiles compared to energy and resource–linked assets.



Risk/Return Comparative Analysis

A comparison of annualised volatility and returns to identify attractiveness of selected stocks.

- **SHP.JO (Return: 21.33%, Risk: 29.84%)** presents the most attractive standalone profile with the highest return and lowest volatility.
- **AGL.JO (Return: 17.78%, Risk: 41.33%)** materially higher risk with solid return potential.
- **PRX.JO (Return: 13.97%, Risk: 40.48%)** elevated risk with moderate growth potential.
- **FSR.JO (Return: 9.61%, Risk: 33.36%)** moderate risk with steady, although lower growth.
- **SOL.JO (Return: 1.91%, Risk: 70.45%)** minimal return with severely high risk. This profile is the least attractive.

Key Observations

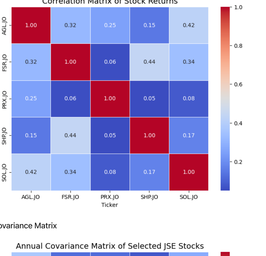
- SHP.JO is the most efficient asset over the selected period with an attractive risk/return ratio.
- SOL.JO increases portfolio risk significantly without a proportionate return contribution.
- Diversification benefits will depend on correlation structure rather than return alone.

Although individual asset performance is informative, an optimal portfolio construction requires analysing correlation and covariance to determine whether diversification can reduce risks and optimise returns.

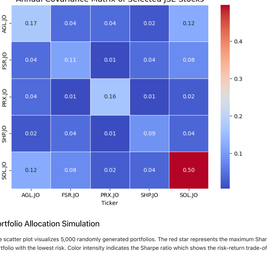
Note: Volatility scales with the square root of the period due to properties of variance under independent returns.

Asset Correlation

Diversification benefits arise when correlations between assets are less than 1.



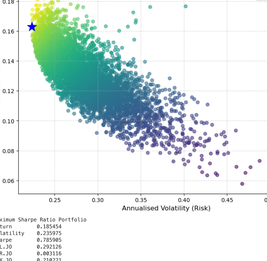
Covariance Matrix



Portfolio Allocation Simulation

The scatter plot visualizes 5,000 randomly generated portfolios. The red star represents the maximum Sharpe Ratio portfolio with the best risk-adjusted return, and the blue star represents the minimum volatility portfolio with the lowest risk. Color intensity indicates the Sharpe ratio which shows the risk-return trade-off and helps investors choose optimal allocations.

The Sharpe ratio measures return per unit of risk, allowing portfolios to be compared on a risk-adjusted basis rather than raw return alone.

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Maximum Sharpe Ratio Portfolio
Return    0.185454
Volatility    0.235975
Sharpe    0.785905
AGL.JO    0.292126
FSR.JO    0.083116
PRX.JO    0.218221
SHP.JO    0.406173
SOL.JO    0.000364
Name: 4313, dtype: float64
Minimum Volatility Portfolio
Return    0.102914
Volatility    0.225918
Sharpe    0.726385
AGL.JO    0.165769
FSR.JO    0.226286
PRX.JO    0.218961
SHP.JO    0.379475
SOL.JO    0.009588
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The simulated portfolio with the maximum Sharpe ratio has an expected annual return of 18.55% with an annualised volatility of 23.60%. The optimal allocation heavily favours SHP.JO (48.61%) and AGL.JO (29.21%), while the remaining stocks have minimal allocation. This portfolio offers the best risk-adjusted return among the 5,000 simulated portfolios, balancing expected returns against volatility.

Conclusion

This analysis evaluated the historical performance of five JSE-listed equities between 2018 and 2025 using modern portfolio theory principles. By comparing annualised returns, volatility, correlations, and simulated portfolio allocations, several key insights emerged.

SHP.JO demonstrated the strongest individual risk–return profile, combining high returns with comparatively low risk. In contrast, SOL.JO exhibited high volatility with very low return contribution, making it less attractive on a risk-adjusted basis.

The Monte Carlo simulation of 5,000 random portfolios revealed that the maximum Sharpe ratio portfolio achieved an expected annual return of approximately 19% with volatility near 24%. The optimal portfolio allocated the majority of capital to SHP.JO, reflecting its superior risk-adjusted characteristics, while assigning minimal weight to higher-volatility, lower-efficiency assets.

This outcome reinforces a core principle of portfolio theory: optimal asset allocation depends not only on individual returns but on covariance structure and diversification benefits. Assets that improve overall portfolio efficiency receive higher weight, even if they are not the absolute highest-return assets.

Limitations

This analysis relies on historical data and assumes return distributions remain stable over time. Market regimes, macroeconomic conditions, and structural changes may alter future performance. Additionally, transaction costs, taxes, and liquidity constraints were not accounted for in the simulation.

Final Insight

The project demonstrates that disciplined quantitative analysis can transform raw market data into actionable portfolio insights. By focusing on risk-adjusted returns and diversification, investors can construct portfolios that are more efficient than any individual asset.